Conjecture 00000001670: incompatible numerical characterizations

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1 The necessary numerical clauses

The English source assigns the same real constant P_3 both values

$$P_3 = 0.918 = \frac{459}{500}, \qquad P_3 = \frac{2}{\pi}\arctan\sqrt{2}.$$

The Chinese source also prints these two numerical characterizations. They cannot hold simultaneously. We disprove this necessary part of the conjecture, hence the stated conjunction, without assuming or formalizing a random-graph limit. We do not claim to determine a genuine max-cut constant, or to disprove the separate asymptotic or concentration clauses.

Even interpreting 0.918 as a three-decimal approximation does not repair the contradiction. We allow the inclusive interval

$$\left|P_3 - \frac{459}{500}\right| \le \frac{1}{2000}, \qquad \text{equivalently} \qquad 0.9175 \le P_3 \le 0.9185.$$

This includes either possible tie at a rounding endpoint.

2 A strict analytic bound

Set $c = (2/\pi)\arctan\sqrt{2}$. The ordinary nonnegative real square root is strictly increasing on positive inputs, so $\sqrt{2} < \sqrt{3}$. The principal real inverse tangent is strictly increasing and satisfies $\arctan\sqrt{3} = \pi/3$. Since $\pi > 0$, multiplication by $2/\pi$ preserves the strict inequality and gives

$$c < \frac{2}{\pi}\arctan\sqrt{3} = \frac{2}{\pi}\frac{\pi}{3} = \frac{2}{3}.$$

Every function and constant in this chain is Mathlib's actual real analytic object in the Lean proof, not an assumed numerical relation.

The rational comparisons now leave a large gap:

$$\frac{459}{500} - c > \frac{459}{500} - \frac{2}{3} = \frac{377}{1500} > \frac{1}{4}.$$

Under the rounding interpretation, $P_3 \ge 367/400$ and therefore

$$P_3 - c > \frac{367}{400} - \frac{2}{3} = \frac{301}{1200} > \frac{1}{4}.$$

Thus the exact closed form differs by more than $1/4$ even from every value in the permitted rounding interval. No numerical approximation to π, $\sqrt{2}$, or $\arctan\sqrt{2}$ is needed.

3 Formal scope and previous submission

The formal definition `closedForm` is literally `(2 / Real.pi) * Real.arctan (Real.sqrt 2)` in $\mathbb{R}$. `Rounded0918` is the absolute-error interval above. The file proves the square-root comparison, inverse-tangent comparison, strict 2/3 bound, and the two strict error bounds before proving

$$\forall P \in \mathbb{R}, \quad \neg\big(P = 459/500 \ \wedge \ P = c\big)$$

and

$$\forall P \in \mathbb{R}, \quad \neg\big(|P - 459/500| \le 1/2000 \ \wedge \ P = c\big).$$

The final two theorems also negate existence of any sequence $P : \mathbb{N} \to \mathbb{R}$ satisfying the respective requirements at index 3. These universally quantified statements apply to any proposed real P_3; no special definition of a graph parameter is inserted to make the contradiction hold.

Closed PR 189, by orionsheep, already presented the same elementary analytic comparison in prose. We acknowledge that prior observation; this submission does not claim it as a new discovery. The maintainer's review rejected its Lean component because it proved only natural-number comparisons and reflexive equalities. The load-bearing real π, square root, inverse tangent, and P_3 contradiction were absent from Lean. The present proof formalizes that entire chain and additionally covers the rounding interpretation. It relies on the existing, proved Mathlib analytic theorems rather than treating the analytic steps as assumptions.

4 Background and verification

Dembo, Montanari and Sen, *Extremal Cuts of Sparse Random Graphs*, study a large-degree asymptotic governed by a constant P_*. Their result provides background for the source's terminology. We do not identify P_* with the source's P_3, use a reported decimal for P_* as a premise, or claim to formalize their probabilistic result. The disproof here is solely the inconsistency of the two printed necessary numerical characterizations.

Lean 4.33.0 and Mathlib commit `db584cd6d46c92f209a44c0f1c829460d327499d` are pinned. The package provides a new Main build, ten transitive axiom audits, and full official `leanchecker --fresh Main` replay, with source-bound commands, exit codes and logs. No `sorry`, `native_decide` or additional axiom is used. The LaTeX source and rendered PDF are included. OpenAI Codex assisted; a separate supervising Codex agent reviewed the formal chain and report. This is AI review, not human certification. Organizer acceptance is pending.